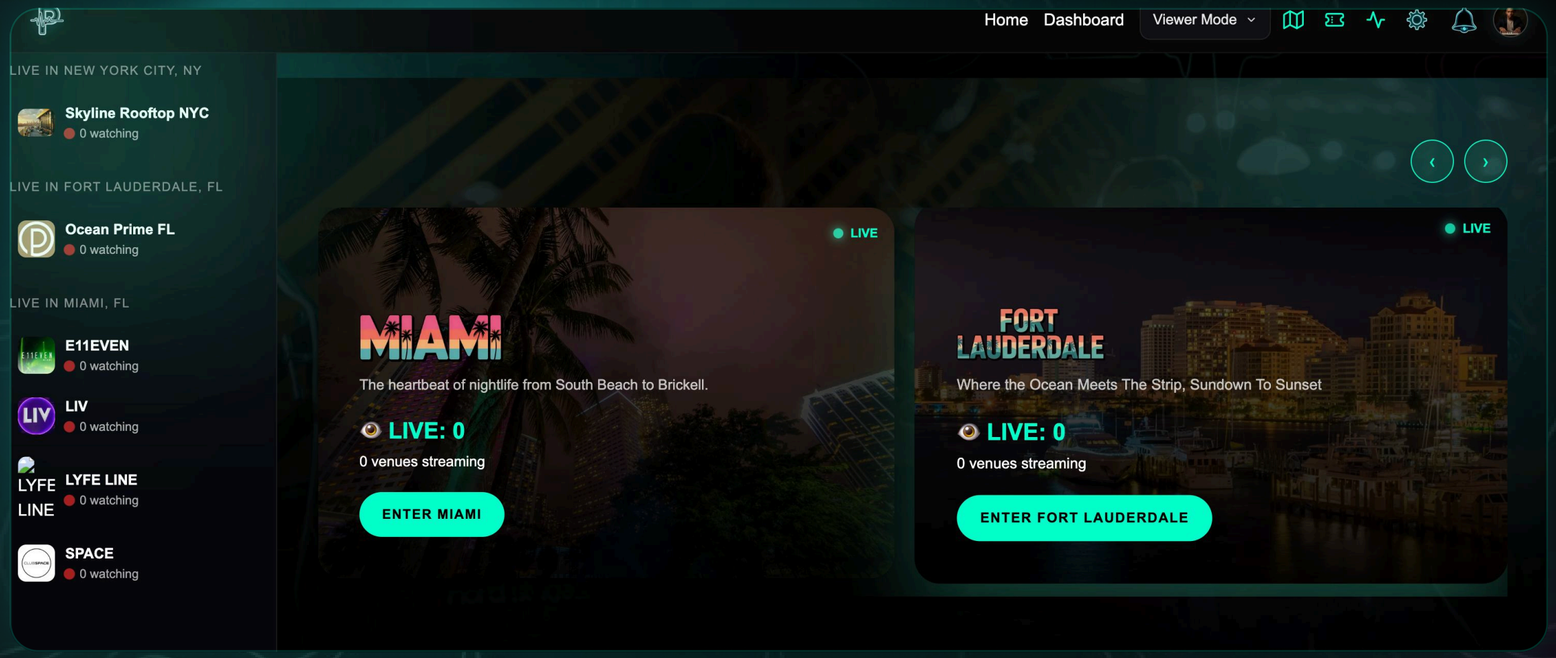
PULSE

Containerized Full-Stack System Supporting Live Venue Streaming, Geospatial Discovery, and Real-Time Engagement Signals



OVERVIEW

Pulse is a real-time streaming and venue engagement platform designed to surface live experiences happening across cities. The platform aggregates live venues, enables creators to stream directly from events, and allows viewers to discover and interact with nightlife in real time. The system combines streaming, geospatial discovery, and live engagement signals to create a dynamic entertainment network.

ENGINEERING CONTRIBUTIONS

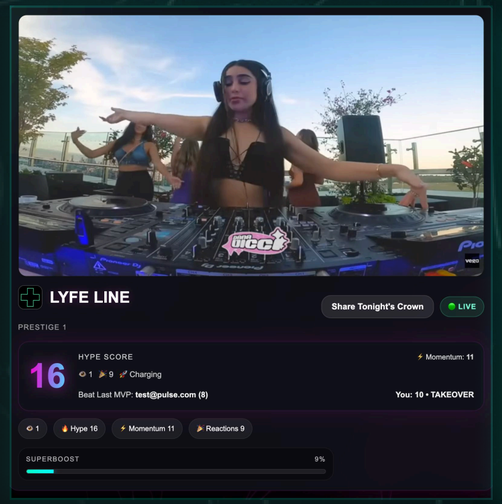
- Led full-stack development of the Pulse real-time streaming platform architecture
- Developed responsive React interfaces for live venue discovery and engagement
- Implemented backend APIs supporting venue, user, and stream management
- Built real-time event data pipelines powering live engagement indicators
- Architected modular services for scalability and maintainability

ARCHITECTURE

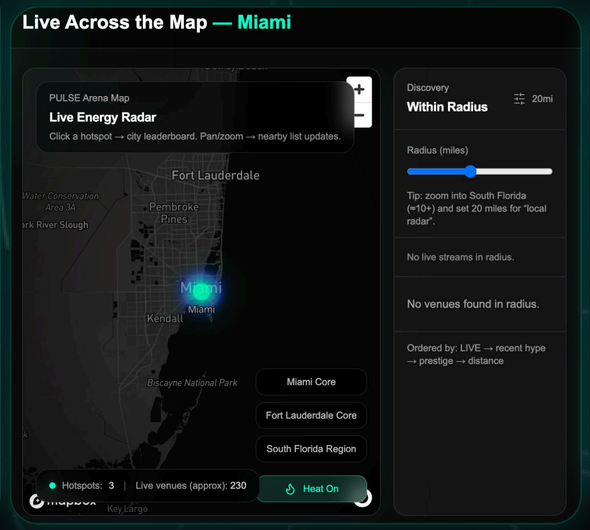
- Containerized services using Docker for consistent deployment
- Decoupled frontend, API, and data services for scalability
- Implemented service boundaries supporting live streaming workflows
- Designed system structure to support high-frequency real-time events

TECHNOLOGY STACK

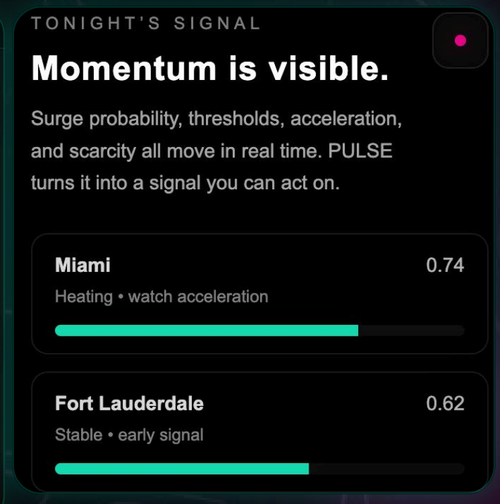
Frontend
React • TypeScript • TailwindCSS • Mapbox
Backend
Node.js • Express • REST APIs
Real-Time Systems
WebSockets • Event Streams
Infrastructure
Docker • Cloud Deployment
Data Layer
PostgreSQL • Redis



Live Venue Stream Interface
Real-Time DJ Broadcast With Engagement Indicators



Geospatial Venue Discovery
Live Venues Mapped By Real-Time Activity Signals



Engagement Signal Dashboard
Momentum Indicators And Real-Time Event Analytics

Personal project — system design, frontend architecture, backend services, and infrastructure.